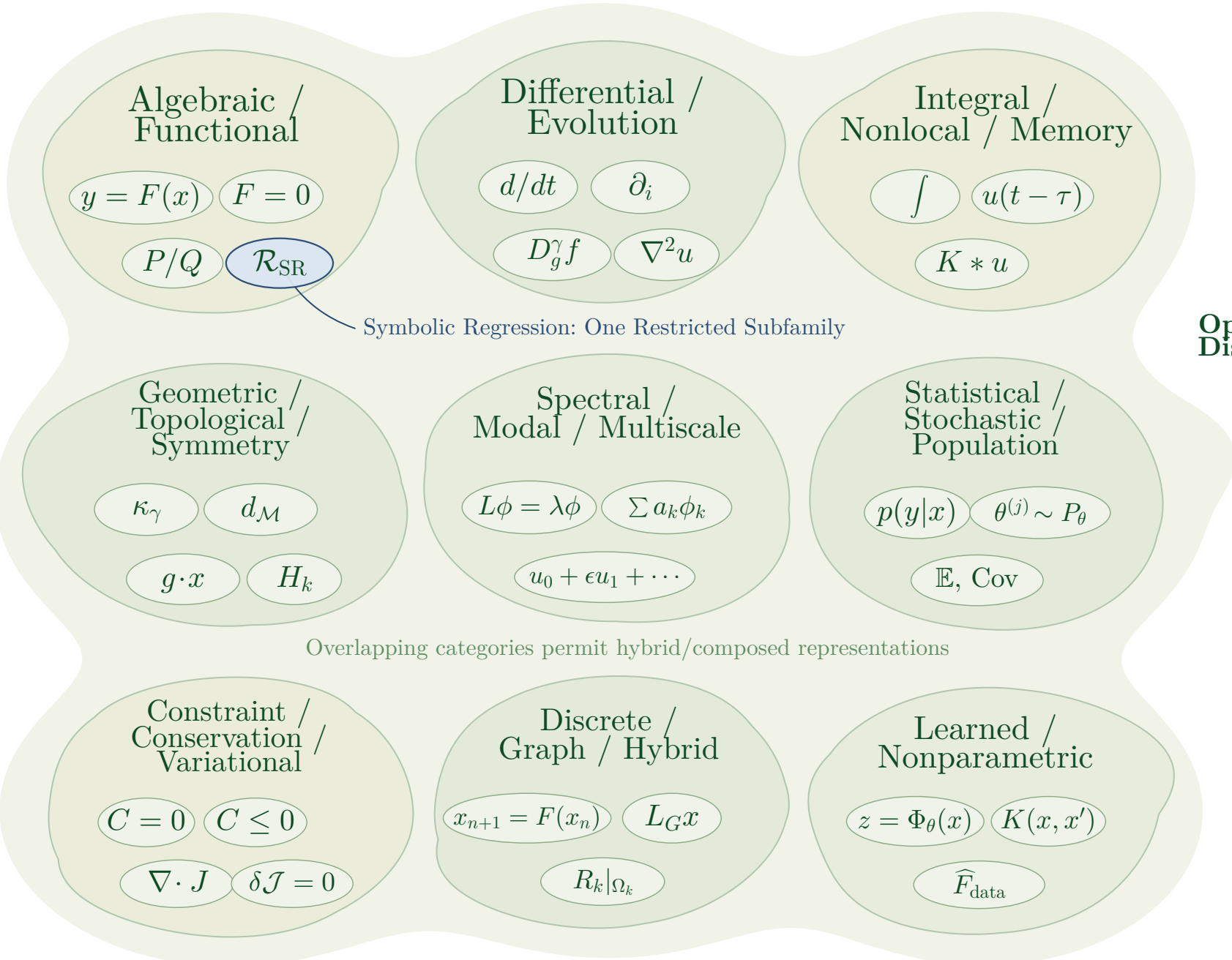


Admissible Relational Space $\mathfrak{R}_q$

Coordinate-Generation Operators

Representative groups; recursively composable



Symbolic Regression: One Restricted Subfamily

Overlapping categories permit hybrid/composed representations

Open, Expandable
Discovery Frontier

$\partial, d, \Delta, D^\gamma$ **Local**
Derivatives, differences,
tangent / phase constructions

$\int, *, \mathcal{L}_\tau$ **Nonlocal / History**
Integrals, convolutions,
lags and memory operators

$\mathcal{T}, \mathcal{B}, \Pi$ **Algebraic / Basis**
Transforms, ratios, reductions,
nondimensional groups and bases

$\mathcal{G}, \mathcal{S}$ **Structural / Statistical**
Geometry, symmetry, graph,
spectral and statistical maps

Φ_θ **Learned**
Embeddings, kernels,
neural / surrogate coordinates

$\mathcal{M}$ **Model Primitives**
Numerical simulators, pretrained
models, scientific codes, agents

$$G = G_{i_k} \circ \dots \circ G_{i_2} \circ G_{i_1}$$

Operators may compose recursively

Key Category Illustrative subfamily

Illustrative, non-exhaustive categories; each SIR run instantiates a finite, auditable grammar.